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MS-AI

QUIZ : MACHINE LEARNING

Date 25/03/2026

$$L=6, S=6$$

1-

Entropy for Full dataset:

$$P(L) = 6/12 = 1/2$$

$$P(S) = 6/12 = 1/2$$

$$H(S) = -6/12 \log_2 \frac{6}{12} - \frac{6}{12} \log_2 \frac{6}{12}$$

$$\Rightarrow -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2}$$

$$\Rightarrow -1/2 (-1) - 1/2 (-1)$$

$$\Rightarrow 0.5 + 0.5$$

$$H(S) \Rightarrow 1$$

ii)

Info. Gain for All Attributes:

For Experience:

$$H(\text{low}) = -4/4 \log_2 4/4 - 0/4 \log_2 0/4 \Rightarrow 0$$

$$H(\text{medium}) = -1/4 \log_2 1/4 - 3/4 \log_2 3/4$$

$$\Rightarrow 0.5 + 0.811 = 0.811$$

$$H(\text{high}) = -1/4 \log_2 1/4 - 3/4 \log_2 3/4$$

$$\Rightarrow 0.811$$

$$\text{Info-Gain} = 4/12 (0) + 4/12 (0.811) + 4/12 (0.811)$$

$$\Rightarrow 0 + 0.270 + 0.270$$

$$\Rightarrow 0.540$$

$$IG(\text{experience}) \Rightarrow 1 - 0.540 = 0.460$$

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Q. For (Job satisfaction):

$$H(\text{low}) = -4/4 \log_2(1) - 0 = 0$$

$$H(\text{medium}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5}$$

$$\Rightarrow 0.5288 + 0.4422$$

$$\Rightarrow 0.971$$

$$H(\text{high}) = -0/3 \log_2 0/3 - 3/3 \log_2 3/3 \Rightarrow 0$$

Info Gain:

$$\Rightarrow 4/12(0) + 5/12(0.971) + 3/12(0)$$

$$\Rightarrow 0 + 0.404 + 0 \Rightarrow 0.404$$

$$IG(\text{Job satn factor}) = 1 - 0.404 \Rightarrow 0.596$$

For Monthly Overtime:

$$H(\text{yes}) = -5/6 \log_2 5/6 - 1/6 \log_2 1/6$$

$$\Rightarrow 0.2192 + 0.4308 = 0.650$$

$$H(\text{No}) = -1/6 \log_2 1/6 - 5/6 \log_2 5/6$$

$$\Rightarrow 0.650$$

Info Gain:

$$\Rightarrow 6/12(0.650) + 6/12(0.650)$$

$$\Rightarrow 0.650$$

$$IG(\text{monthly Overtime}) = 1 - 0.650$$

$$\Rightarrow 0.350$$

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For Performance Rating,

$$H(\text{Avg}) = -4/4 \log_2(1) = 0$$

$$H(\text{Good}) = -2/4 \log_2 2/4 - 2/4 \log_2 2/4$$

$$\Rightarrow 1$$

$$H(\text{excellent}) = -4/4 \log_2(1) = 0$$

$$\text{Info Gain} \Rightarrow 4/12(0) + 4/12(1) + 4/12(0)$$

$$\Rightarrow 0.333$$

$$IG(\text{Performance Rating}) = 1 - 0.333 \Rightarrow 0.667$$

(iv)

Comparing All and Selecting the best one:

★ In performance rating we have ~~got~~ got 0.667 which is the best one, so we will select it as root node because it has the highest Information Gain ~~means~~, which means it gives us the most reduction about whether an employee will leave or stay.

(iv)

Construct the DT:

For Avg $\rightarrow 1, 5, 7, 11 \rightarrow$ All L, so it is pure leaf $\Rightarrow$ Leave

For Excellent $\rightarrow 4, 6, 10, 12 \rightarrow$ All S, pure leaf $\Rightarrow$ Stay

For Good $\rightarrow 2, 3, 8, 9 \rightarrow$ L, S, S, L (2L, 2S) $\rightarrow$

Not pure so it needs further splitting

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ID	Attraction	Exp	Job-S	Monthly Overcome
2	L	Low	med	Yes
3	S	med	high	No
8	S	med	med	No
9	L	high	Low	Yes

$$H(\text{Good subset}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \\ \rightarrow 1.00$$

Now if we try Monthly Overtime on Good subset

for Yes $\Rightarrow 2, 9 \rightarrow 2, L \rightarrow 2L, OS$

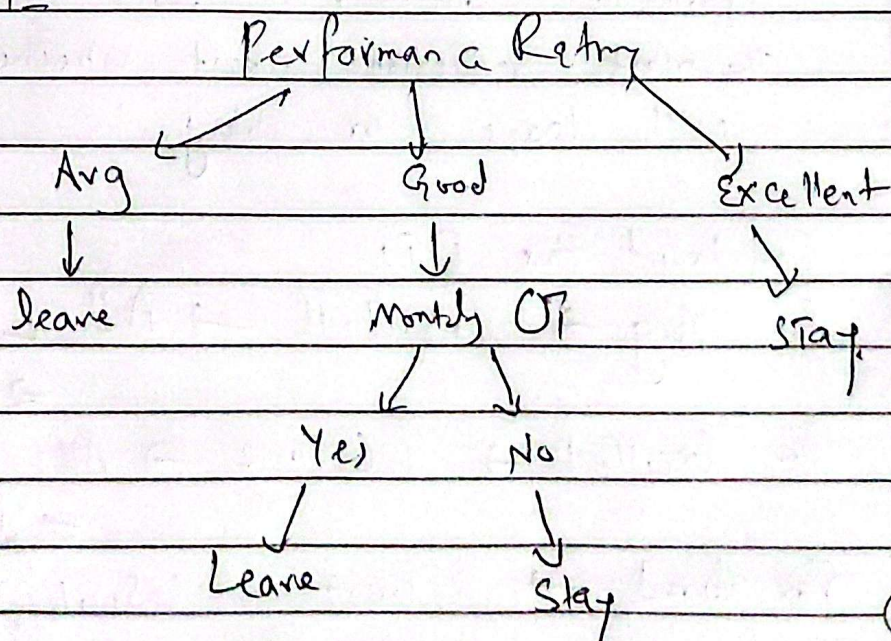
$$H = 0$$

for No $\Rightarrow 3, 8 \rightarrow S, S \rightarrow OS, 2S \Rightarrow H = 0$

$$IG(\text{Good monthly OT / Good}) = 1 - \frac{2}{4}(0) - \frac{2}{4}(0) = 1$$

hence both are perfectly pure ~~subsets~~

Final DT:



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Part B:

A

Employee (X) = (Med, Low, Yes, Good)

Path:

- a. performance Rating = Good $\rightarrow$ Go to Monthly OT
- b. Monthly OT = Yes $\rightarrow$ Leave

So ~~Exit~~ Employee X $\rightarrow$ Leave (L)

A

Employee (Y) = (High, high, No, Excellent)

Path:

- a. Performance Rating = Excellent $\rightarrow$ Pure Leaf
- So Employee ~~(X)~~ Y $\rightarrow$ Stay (S)

A

Employee (Z) = (low, medium, No, Avg)

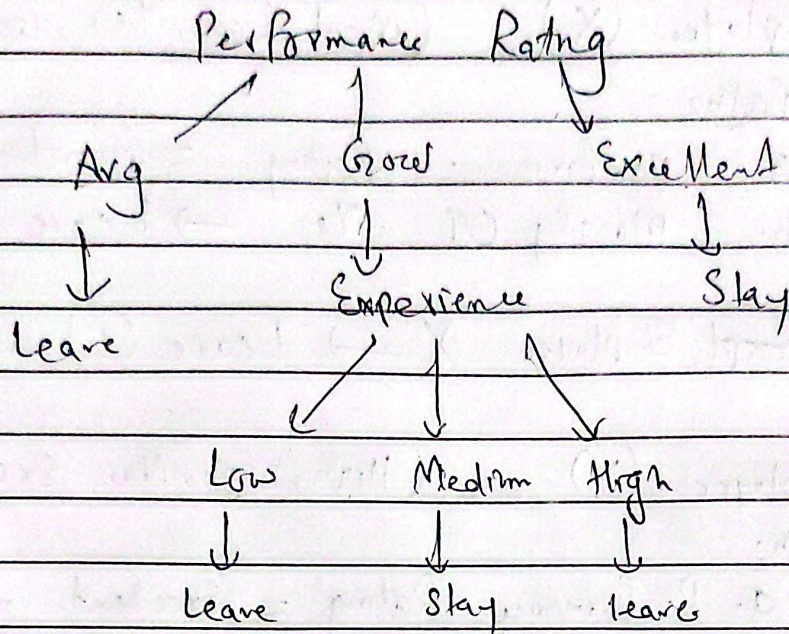
Path:

- a. Performance Rating = Avg $\rightarrow$ Pure Leaf

So employee Z $\rightarrow$ leave (L)

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Final DF
2 =



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